WEEK 9: SALES AND CUSTOMER ANALYTICS SYSTEM

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Scenario:

The Art Workshop wants to analyze its sales data to understand customer purchasing behavior, product performance,and revenue trends. The management needs reports to identify top customers,best-selling products, and profitable categories.

1.Apply SQL Aggregate Functions:

SQL Aggregate functions perform calculations on a set of multiple records and return a single summarized value or a result for each group.

COUNT() – The COUNT() function counts the number of records in a table.
SUM() – The SUM() function calculates the total of a numeric column.
AVG() – The AVG() function calculates the average values of a numeric column.
MAX() – The MAX() function returns the highest value in a column.
MIN() – The MIN() function returns the lowest value in a column.

2.Generate Total Sales Reports:

Tasks: Write SQL queries to calculate:
1.Total number of orders placed

```
SQL> SELECT COUNT(*) FROM ORDERS;

  COUNT(*)
----------
        10
```

```
SQL> SELECT COUNT(*) AS TOTAL_NO_OF_ORDERS
  2  FROM Orders;

TOTAL_NO_OF_ORDERS
------------------
                10
```

2.Total revenue generated from sales

```
SQL> SELECT SUM(TOTAL_AMOUNT)FROM ORDERS;

SUM(TOTAL_AMOUNT)
-----------------
           431800

SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_REVENUVE FROM ORDERS;

TOTAL_REVENUVE
--------------
        431800
```

3.Average order value

```
SQL> SELECT AVG(TOTAL_AMOUNT) AS ORDER_VALUE FROM ORDERS;

ORDER_VALUE
-----------
      43180
```

➔ Average order value of each order at a time

```
SQL> SELECT ORDER_ID,AVG(TOTAL_AMOUNT)FROM ORDERS GROUP BY ORDER_ID;

  ORDER_ID AVG(TOTAL_AMOUNT)
---------- ----------------
      1001            70000
      1002            45000
      1003             3400
      1004            75000
      1006            55000
      1007            67500
      1008            45000
      1009             3400
      1010             7500
      1011            60000

10 rows selected.
```

4.Highest and lowest order amount

```
SQL> SELECT MIN(TOTAL_AMOUNT) FROM ORDERS;

MIN(TOTAL_AMOUNT)
-----------------
             3400

SQL> SELECT MAX(TOTAL_AMOUNT) FROM ORDERS;

MAX(TOTAL_AMOUNT)
-----------------
            75000

SQL> SELECT MIN(TOTAL_AMOUNT) AS LOWEST_ORDERS FROM ORDERS;

LOWEST_ORDERS
-------------
         3400

SQL> SELECT MAX(TOTAL_AMOUNT) AS HIGHEST_ORDERS FROM ORDERS;

HIGHEST_ORDERS
--------------
         75000
```

5.Total sales for a specific period

```
SQL> SELECT COUNT(ORDER_ID) AS TOTAL_ORDERS,SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE,AVG(TOTAL_AMOUNT) AS AVG_ORDER_VALUE FROM ORDERS;

TOTAL_ORDERS TOTAL_REVENUE AVG_ORDER_VALUE
------------ ------------- ---------------
          10        431800           43180

SQL>
```

3.Customer Purchase Analysis:

1.Total number of orders placed by each customer.

```
SQL>  SELECT COUNT(*),CUSTOMER_ID FROM ORDERS GROUP BY CUSTOMER_ID;

  COUNT(*) CUSTOMER_ID
---------- --------------------
         2 C-1001
         2 C-1004
         2 C-1002
         2 C-1003
         2 C-1005
```

2.Total amount spent by each customer.

```
SQL> SELECT CUSTOMER_ID, SUM(TOTAL_AMOUNT)
  2  FROM ORDERS
  3  GROUP BY CUSTOMER_ID;

CUSTOMER_ID          SUM(TOTAL_AMOUNT)
-------------------- -----------------
C-1001                          125000
C-1004                           78400
C-1002                          112500
C-1003                           48400
C-1005                           67500
```

3.Average spending per customer.

```
SQL> SELECT CUSTOMER_ID, AVG(TOTAL_AMOUNT)
  2  FROM ORDERS
  3  GROUP BY CUSTOMER_ID;

CUSTOMER_ID          AVG(TOTAL_AMOUNT)
-------------------- -----------------
C-1001                           62500
C-1004                           39200
C-1002                           56250
C-1003                           24200
C-1005                           33750
```

4.Customers with maximum purchase amount.

```
SQL> SELECT CUSTOMER_ID,TOTAL_AMOUNT AS MAX_PURCHASE_AMOUNT FROM ORDERS WHERE TOTAL_AMOUNT=(SELECT MAX(TOTAL_AMOUNT) FROM ORDERS);

CUSTOMER_ID          MAX_PURCHASE_AMOUNT
-------------------- -------------------
C-1004                             75000
```

5.Customers with fewer purchases.

```
SQL> SELECT CUSTOMER_ID,TOTAL_AMOUNT AS MIN_PURCHASE_AMOUNT FROM ORDERS WHER
E TOTAL_AMOUNT=(SELECT MIN(TOTAL_AMOUNT)FROM ORDERS);

CUSTOMER_ID          MIN_PURCHASE_AMOUNT
-------------------- -------------------
C-1003                              3400
C-1004                              3400
```

4.Find Top Customers Based on Purchase Amount:

1.Top 5 customers based on total spending

```
SQL> SELECT CUSTOMER_ID, SUM(TOTAL_AMOUNT)
  2  FROM ORDERS
  3  GROUP BY CUSTOMER_ID
  4  ORDER BY SUM(TOTAL_AMOUNT) DESC;

CUSTOMER_ID          SUM(TOTAL_AMOUNT)
-------------------- -----------------
C-1001                          125000
C-1002                          112500
C-1004                           78400
C-1005                           67500
C-1003                           48400
```

2.Customers with maximum number of orders

```
SQL> SELECT COUNT(*) AS NO_OF_ORDERS, CUSTOMER_ID
  2  FROM ORDERS
  3  GROUP BY CUSTOMER_ID
  4  ORDER BY NO_OF_ORDERS DESC;

NO_OF_ORDERS CUSTOMER_ID
------------ --------------------
           2 C-1001
           2 C-1004
           2 C-1005
           2 C-1003
           2 C-1002
```

3.Frequent customers

```
SQL> SELECT CUSTOMER_ID,COUNT(*) AS NO_OF_ORDERS FROM ORDERS
  2  GROUP BY CUSTOMER_ID ORDER BY NO_OF_ORDERS DESC;

CUSTOMER_ID          NO_OF_ORDERS
-------------------- ------------
C-1001                          2
C-1004                          2
C-1005                          2
C-1003                          2
C-1002                          2
```

4.High-value customers

```
SQL> SELECT CUSTOMER_ID,SUM(TOTAL_AMOUNT) AS TOTAL_SPENDING
  2  FROM ORDERS
  3  GROUP BY CUSTOMER_ID
  4  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_ID          TOTAL_SPENDING
-------------------- --------------
C-1001                       125000
C-1002                       112500
C-1004                        78400
C-1005                        67500
C-1003                        48400
```

5. Identify Best-Selling Products:

1.Products with maximum sales quantity

```
SQL> SELECT PRODUCT_NAME,QUANTITY FROM PRODUCT ORDER BY QUANTITY DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY
----------
Croptops and shirts
         2

Laptops
         2

Fish tank
         2


PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY
----------
Tiles
         1

Unruled notes
         1

Television
         1


PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY
----------
Mixer Grinder
         0
```

```
Mobile Charger
         0

Pens
         0


PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY
----------
Notebooks
         0

Jeans
         0

T-Shirts
         0


PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY
----------
Rice Bag
         0

Headphones
         0

Keyboard
         0


15 rows selected.
```

2.Products generating highest revenue

```
SQL> SELECT PRODUCT_NAME, PRODUCT_ID,
  2         SUM(QUANTITY * PRICE) AS REVENUE
  3  FROM PRODUCT
  4  GROUP BY PRODUCT_NAME, PRODUCT_ID
  5  ORDER BY REVENUE DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT    REVENUE
------- ----------
Laptops
P1001       135000

Croptops and shirts
P1002        90000

Television
P1004        75000


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT    REVENUE
------- ----------
Tiles
P1006        55000

Unruled notes
P1005         7500

Fish tank
P1003         6800


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT    REVENUE
```

```
Jeans
P1010             0

Mixer Grinder
P1013             0


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT    REVENUE
------- ----------
Rice Bag
P1015             0

Keyboard
P1008             0

Mobile Charger
P1014             0


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT    REVENUE
------- ----------
Headphones
P1007             0

Notebooks
P1011             0

T-Shirts
P1009             0


15 rows selected.
```

3.Least-selling products

```
SQL> SELECT PRODUCT_NAME, PRODUCT_ID,SUM(PRICE*QUANTITY)AS SELLING_PRICE
  2  FROM PRODUCT GROUP BY PRODUCT_NAME, PRODUCT_ID
  3   ORDER BY SELLING_PRICE ASC;

PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT SELLING_PRICE
------- -------------
Headphones
P1007               0

Keyboard
P1008               0

Pens
P1012               0


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT SELLING_PRICE
------- -------------
Notebooks
P1011               0

T-Shirts
P1009               0

Mixer Grinder
P1013               0


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT SELLING_PRICE
------- -------------
Mobile Charger
```

```
Mobile Charger
P1014                   0

Jeans
P1010                   0

Rice Bag
P1015                   0


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT SELLING_PRICE
------- -------------
Fish tank
P1003                6800

Unruled notes
P1005                7500

Tiles
P1006               55000


PRODUCT_NAME
--------------------------------------------------------------------------------
PRODUCT SELLING_PRICE
------- -------------
Television
P1004               75000

Croptops and shirts
P1002               90000

Laptops
P1001              135000
```

4.Products requiring promotion

```
SQL> SELECT OD.PRODUCT_ID, COUNT(*) AS NO_OF_SALES
  2  FROM ORDERS O, ORDER_DETAILS OD
  3  WHERE O.ORDER_ID = OD.ORDER_ID
  4  AND O.ORDER_STATUS = 'Delivered'
  5  GROUP BY OD.PRODUCT_ID
  6  ORDER BY NO_OF_SALES ASC;

PRODUCT NO_OF_SALES
------- -----------
P1006             1
P1003             2
```

6.Perform Category-wise Sales Analysis:

1.Total sales generated by each category.

```
SQL> SELECT SUM(O.TOTAL_AMOUNT) AS TOTAL_SALE,
  2         P.PRODUCT_NAME
  3  FROM ORDERS O
  4  JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
  5  JOIN PRODUCT P ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_NAME;

TOTAL_SALE
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
    137500
Laptops

     90000
Croptops and shirts

      6800
Fish tank


TOTAL_SALE
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
      7500
Unruled notes

     75000
Television

     55000
Tiles


6 rows selected.
```

2.Number of Products sold in each category.

```
SQL> SELECT C.CATEGORY_NAME, COUNT(OD.PRODUCT_ID) AS TOTAL_PRODUCTS_SOLD
  2  FROM CATEGORY C
  3  JOIN PRODUCT P ON C.CATEGORY_ID = P.CATEGORY_ID
  4  JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  5  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
Clothing
                   2

Stationary
                   2

Home Appliances
                   1


CATEGORY_NAME
--------------------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD
-------------------
Mobile Parts
                   1

Electronics
                   2

Provisions
                   1


6 rows selected.
```

3.Highest revenue-generating category.

```
SQL> SELECT SUM(O.TOTAL_AMOUNT) AS TOTAL_SALE,
  2         P.PRODUCT_NAME
  3  FROM ORDERS O
  4  JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
  5  JOIN PRODUCT P ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_NAME
  7  ORDER BY TOTAL_SALE DESC;

TOTAL_SALE
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
    137500
Laptops

     90000
Croptops and shirts

     75000
Television


TOTAL_SALE
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     55000
Tiles

      7500
Unruled notes

      6800
Fish tank


6 rows selected.
```

4.Average sales per category.

```
SQL> SELECT AVG(O.TOTAL_AMOUNT) AS AVG_SALE,
  2     P.PRODUCT_NAME
  3  FROM ORDERS O
  4  JOIN ORDER_DETAILS OD ON O.ORDER_ID = OD.ORDER_ID
  5  JOIN PRODUCT P ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_NAME;

  AVG_SALE
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     68750
Laptops

     45000
Croptops and shirts

      3400
Fish tank


  AVG_SALE
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
      7500
Unruled notes

     75000
Television

     55000
Tiles

6 rows selected.
```

7.Generate Business Analytics Reports:

Report 1: Sales Performance Report

- Total sales
- Number of orders
- Average order value
- Highest order value

```
SQL> SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  2         COUNT(ORDER_ID) AS NUMBER_OF_ORDERS,
  3         AVG(TOTAL_AMOUNT) AS AVERAGE_ORDER_VALUE,
  4         MAX(TOTAL_AMOUNT) AS HIGHEST_ORDER_VALUE
  5  FROM ORDERS;

TOTAL_SALES NUMBER_OF_ORDERS AVERAGE_ORDER_VALUE HIGHEST_ORDER_VALUE
----------- ---------------- ------------------- -------------------
     431800               10               43180               75000
```

Report 2: Customer Analytics Report:

- Customer name
- Number of orders
- Total spending
- Purchase frequency

```
SQL> SELECT C.*,
  2         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS,
  3         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  4  FROM CUSTOMER C
  5  JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  6  GROUP BY C.CUSTOMER_ID, C.FIRST_NAME, C.LAST_NAME,
  7           C.EMAIL, C.PHONE, C.CREATED_AT;

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A NUMBER_OF_ORDERS
------------------------------ ---------- --------- ----------------
TOTAL_SPENDING
--------------
C-1003               Arun                 kumar
arun200@edss.com               9876543212 15-OCT-24                2
         48400

C-1002               priya                sakthi
priyasakthi@edss.com           9856734456 23-SEP-24                2
        112500

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A NUMBER_OF_ORDERS
------------------------------ ---------- --------- ----------------
TOTAL_SPENDING
--------------

C-1001               Beema                ponting
beema200@edss.com              9234567856 14-FEB-23                2
        125000

C-1005               Reena                das
reenadas200@edss.com           9087654567 30-APR-22                2

CUSTOMER_ID          FIRST_NAME           LAST_NAME
-------------------- -------------------- --------------------
EMAIL                               PHONE CREATED_A NUMBER_OF_ORDERS
------------------------------ ---------- --------- ----------------
TOTAL_SPENDING
--------------
         67500

C-1004               Anu                  Priya
anupriya200@edss.com           8765943256 23-MAY-25                2
         78400
```

Report 3: Product Performance Report:

- Product name
- Quantity sold
- Revenue generated

```
SQL> SELECT P.PRODUCT_NAME,
  2         SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  3         SUM(OD.QUANTITY * OD.PRICE) AS REVENUE_GENERATED
  4  FROM PRODUCT P
  5  JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_NAME;

PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Laptops
            2            135000

Croptops and shirts
            2             90000

Fish tank
            2              6800


PRODUCT_NAME
--------------------------------------------------------------------------------
QUANTITY_SOLD REVENUE_GENERATED
------------- -----------------
Unruled notes
            1              7500

Television
            1             75000

Tiles
            1             55000


6 rows selected.
```

Report 4: Category Analytics Report:

- Category name
- Total products sold
- Total revenue

```
SQL> SELECT C.CATEGORY_NAME,
  2         SUM(OD.QUANTITY) AS TOTAL_PRODUCTS_SOLD,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM CATEGORY C
  5  JOIN PRODUCT P ON C.CATEGORY_ID = P.CATEGORY_ID
  6  JOIN ORDER_DETAILS OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  7  GROUP BY C.CATEGORY_NAME;

CATEGORY_NAME
--------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Clothing
                  2         90000

Stationary
                  2          6800

Home Appliances
                  1         75000


CATEGORY_NAME
--------------------------------------------------------------------
TOTAL_PRODUCTS_SOLD TOTAL_REVENUE
------------------- -------------
Mobile Parts
                  1          7500

Electronics
                  2        135000

Provisions
                  1         55000


6 rows selected.
```